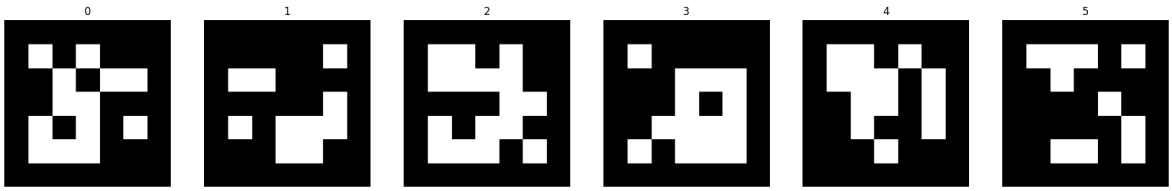
產生ArUco marker

下面範例產生ArUco marker影像，其中DICT_5X5_100表示Marker為5X5的黑白Bit-pattern，一共可有100個，編號0-99號。

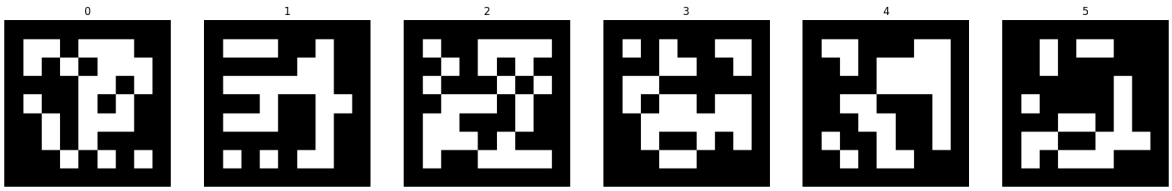
In [1]:
```
import cv2
import cv2.aruco as aruco
import matplotlib.pyplot as plt
arucoDict   = aruco.getPredefinedDictionary(aruco.DICT_5X5_100)
plt.figure(figsize=(24,4))
for i in range(6):
    plt.subplot(1,6,i+1)
    plt.title(i)
    marker_image = aruco.generateImageMarker(arucoDict,i,200)
    plt.imshow(marker_image,cmap='gray')
    plt.axis(False)
plt.show()
```



下面範例產生ArUco marker影像，其中DICT_7X7_50表示Marker為7X7的黑白Bit-pattern，一共可有50個，編號0-49號。

In [2]:
```
import cv2
import cv2.aruco as aruco
import matplotlib.pyplot as plt
arucoDict   = aruco.getPredefinedDictionary(aruco.DICT_7X7_50)

plt.figure(figsize=(24,4))
for i in range(0,6):
    plt.subplot(1,6,1+i)
    plt.title(i)
    marker_image = aruco.generateImageMarker(arucoDict,i,200)
    plt.imshow(marker_image,cmap='gray')
    plt.axis(False)
plt.show()
```



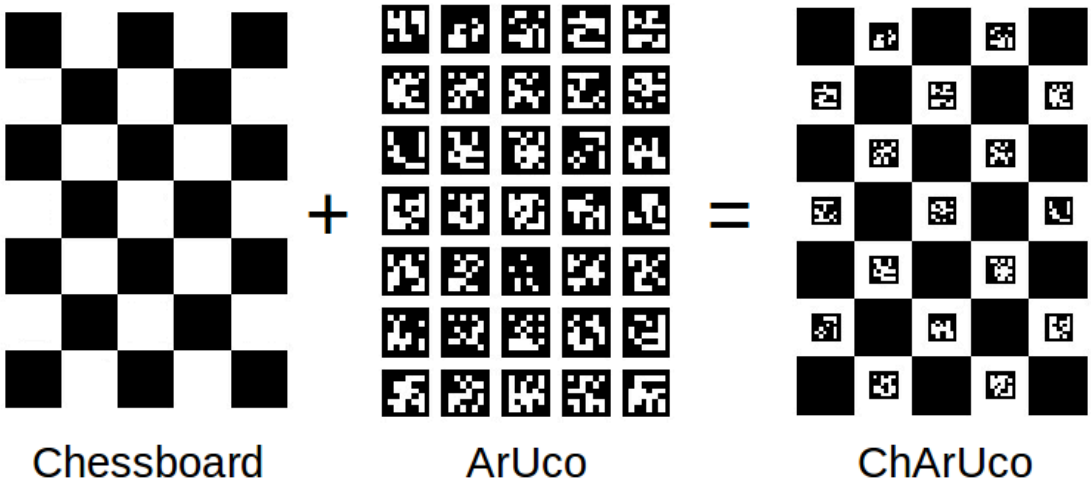
到這裡或是用下面程式，產生特定尺寸(6cmX6cm)的ArUco Marker。

In [3]:
```
import cv2
import cv2.aruco as aruco
import matplotlib.pyplot as plt
arucoDict     = aruco.getPredefinedDictionary(aruco.DICT_7X7_50)
markerSize    = 6 # cm
```

```
markerId       = 6
plt.figure(figsize=(markerSize/2.54,markerSize/2.54))
marker_image = aruco.generateImageMarker(arucoDict,markerId,200)
plt.grid(False)
plt.axis(False)
plt.tight_layout()
plt.imshow(marker_image,cmap='gray')
plt.savefig('aruco_marker.svg')
plt.show()
```

產生ChArUco Marker

什麼是ChArUco board?詳細說明看這裡?



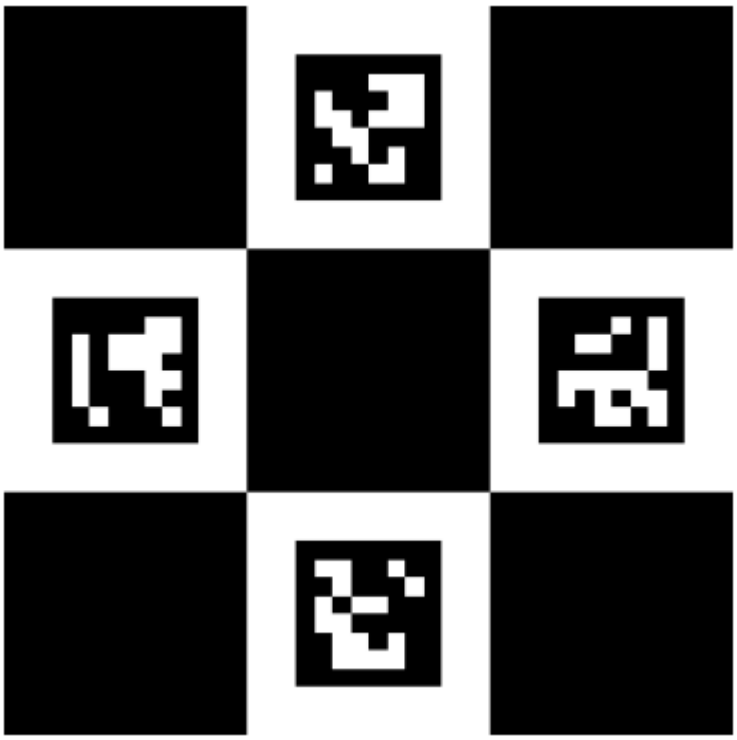
In [4]:
```
#help(aruco.CharucoBoard_create)
```

In [5]:
```
dir(aruco)
```

Out[5]:
```
['ArucoDetector',
 'Board',
 'CORNER_REFINE_APRILTAG',
 'CORNER_REFINE_CONTOUR',
 'CORNER_REFINE_NONE',
 'CORNER_REFINE_SUBPIX',
 'CharucoBoard',
 'CharucoDetector',
 'CharucoParameters',
 'DICT_4X4_100',
 'DICT_4X4_1000',
 'DICT_4X4_250',
 'DICT_4X4_50',
 'DICT_5X5_100',
 'DICT_5X5_1000',
 'DICT_5X5_250',
 'DICT_5X5_50',
 'DICT_6X6_100',
 'DICT_6X6_1000',
 'DICT_6X6_250',
 'DICT_6X6_50',
 'DICT_7X7_100',
 'DICT_7X7_1000',
 'DICT_7X7_250',
 'DICT_7X7_50',
 'DICT_APRILTAG_16H5',
 'DICT_APRILTAG_16h5',
 'DICT_APRILTAG_25H9',
 'DICT_APRILTAG_25h9',
 'DICT_APRILTAG_36H10',
 'DICT_APRILTAG_36H11',
 'DICT_APRILTAG_36h10',
 'DICT_APRILTAG_36h11',
 'DICT_ARUCO_MIP_36H12',
 'DICT_ARUCO_MIP_36h12',
 'DICT_ARUCO_ORIGINAL',
 'DetectorParameters',
 'Dictionary',
 'Dictionary_getBitsFromByteList',
 'Dictionary_getByteListFromBits',
 'GridBoard',
 'RefineParameters',
 '__doc__',
 '__file__',
 '__loader__',
 '__name__',
 '__package__',
 '__path__',
 '__spec__',
 '_native',
 'drawDetectedCornersCharuco',
 'drawDetectedDiamonds',
 'drawDetectedMarkers',
 'extendDictionary',
 'generateImageMarker',
 'getPredefinedDictionary']
```

In [6]:
```
import numpy as np
arucoDict = aruco.getPredefinedDictionary(aruco.DICT_6X6_250)
arucoIds  = np.array([13,29,47,97])
```

```
squareLength= 200
markerLength= 120
markerSize  = 10 # cm
charucoBoard = aruco.CharucoBoard((3,3),squareLength,markerLength,arucoDict)
markerImage = charucoBoard.generateImage((3*100,3*100),0,0) # Square 100x100 pix
#markerImage = aruco.drawCharucoDiamond(arucoDict,arucoIds,squareLength,markerLe
plt.figure(figsize=(markerSize/2.54,markerSize/2.54))
plt.axis(False)
plt.grid(False)
plt.tight_layout()
plt.imshow(markerImage,cmap='gray')
plt.savefig('ChArUco_marker.svg')
plt.show()
```

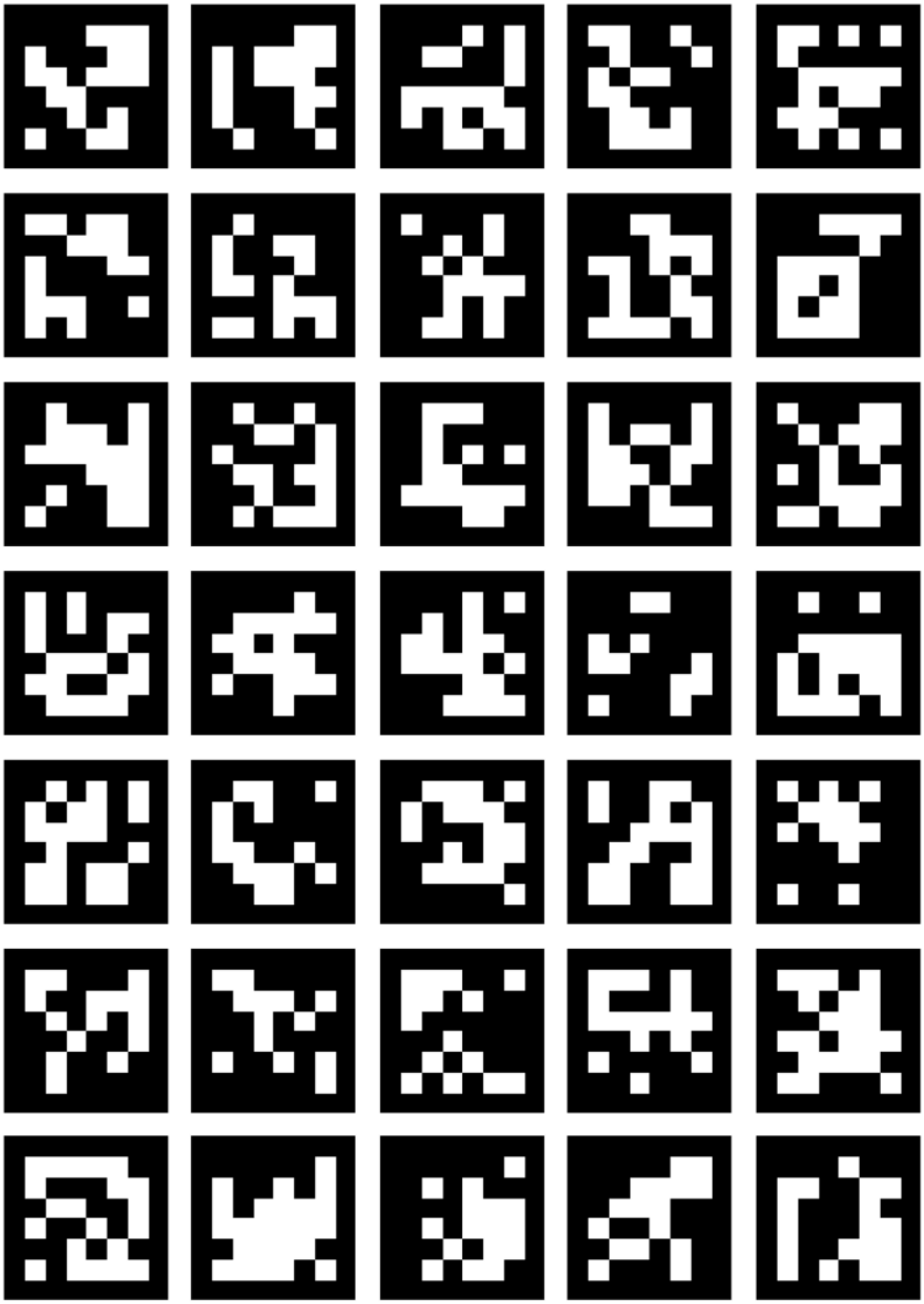


使用ChArUco或ArUco Marker做相機校正

產生ArUco Board。

In [8]:
```
import numpy as np
arucoDict     = aruco.getPredefinedDictionary(aruco.DICT_6X6_250)
gridX         = 5
gridY         = 7
markerSize    = 3.5 #cm
markerGap     = 0.5 #cm
arucoBoard    = aruco.GridBoard((gridX,gridY),markerSize,markerGap,arucoDict) # m
boardImage    = arucoBoard.generateImage((gridX*100,gridY*100),marginSize=10) #5*
boardHeight   = gridY*markerSize+(gridY+1)*markerGap #cm
boardWidth    = gridX*markerSize+(gridX+1)*markerGap
plt.figure(figsize=(boardWidth/2.54,boardHeight/2.54))
plt.grid(False)
plt.axis(False)
plt.tight_layout()
plt.imshow(boardImage,cmap='gray')
```

```
plt.savefig('ArUco_board.svg')
plt.show()
```

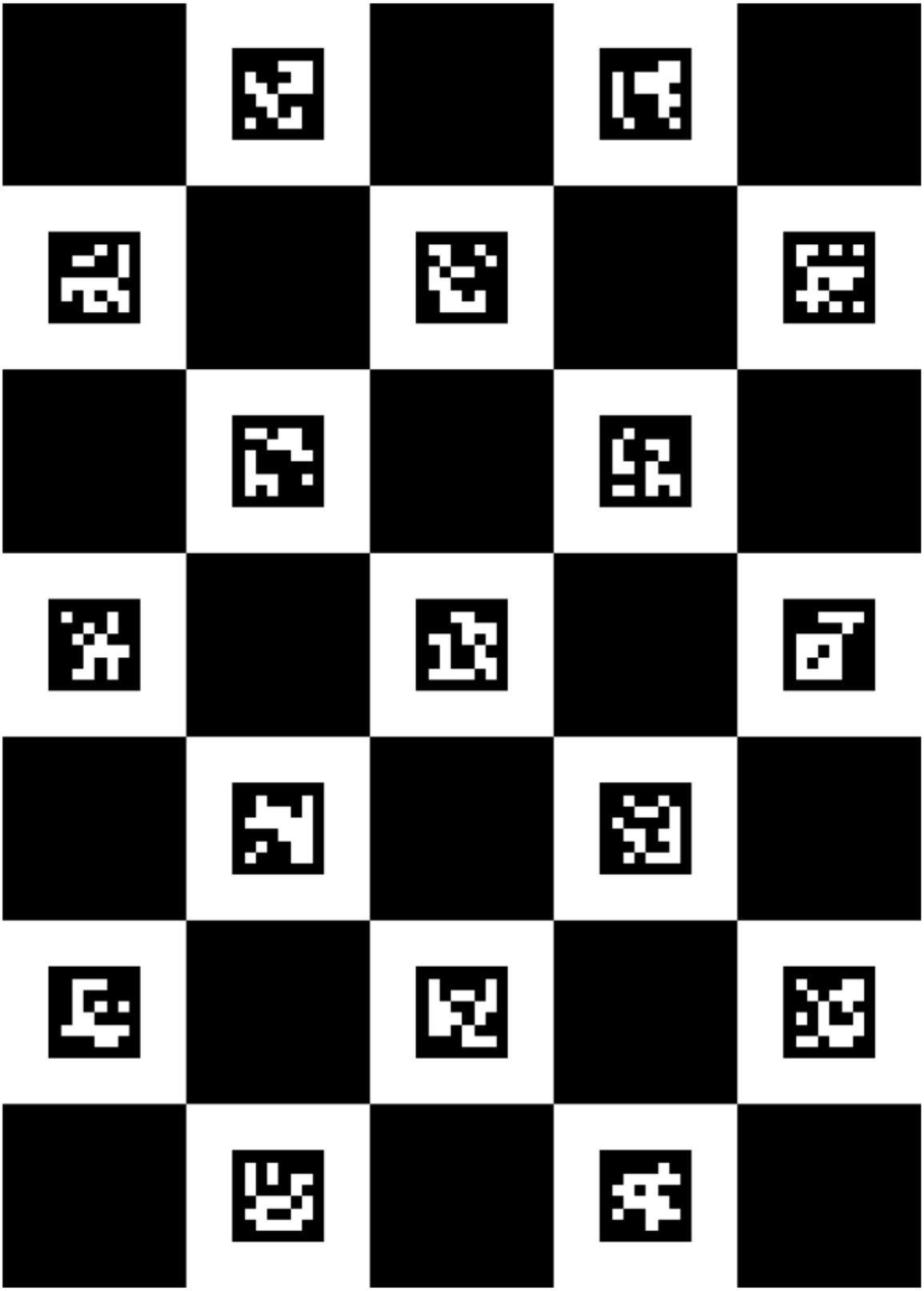


產生ChArUco Board。

根據官方說明，偵測ChArUco board比較準。

In [9]:
```
import numpy as np
arucoDict     = aruco.getPredefinedDictionary(aruco.DICT_6X6_250)
gridX         = 5
```

```
gridY        = 7
squareSize   = 4 #cm
charucoBoard = aruco.CharucoBoard((gridX,gridY),squareSize,squareSize/2,arucoDic
boardImage   = charucoBoard.generateImage((gridX*100,gridY*100),0,0) # Square 10
plt.figure(figsize=(gridX*squareSize/2.54,gridY*squareSize/2.54)) #Board 5*4cm X
plt.grid(False)
plt.axis(False)
plt.tight_layout()
plt.imshow(boardImage,cmap='gray')
plt.savefig('ChArUco_board.svg')
plt.show()
```



偵測ArUco Marker

使用前先準備有關aruco影像處理與marker編碼的參數。有必要，可以調整影像處理的參數arucoParams:

```
arucoParams = aruco.DetectorParameters()
arucoDict   = aruco.getPredefinedDictionary(aruco.DICT_7X7_50)
detector    = aruco.ArucoDetector(arucoDict, arucoParams)
```

然後呼叫偵測aruco marker函式:

```
(corners, ids, rejected) = detector.detectMarkers(frame)
```

若要顯示偵測結果，可以呼叫drawDetectedMarkers

```
aruco.drawDetectedMarkers(frame, corners, ids)
```

corners依照順時針排，但是沒說哪一個角是第一個點。

In [10]:
```
dir(aruco)
```

Out[10]:
```
['ArucoDetector',
 'Board',
 'CORNER_REFINE_APRILTAG',
 'CORNER_REFINE_CONTOUR',
 'CORNER_REFINE_NONE',
 'CORNER_REFINE_SUBPIX',
 'CharucoBoard',
 'CharucoDetector',
 'CharucoParameters',
 'DICT_4X4_100',
 'DICT_4X4_1000',
 'DICT_4X4_250',
 'DICT_4X4_50',
 'DICT_5X5_100',
 'DICT_5X5_1000',
 'DICT_5X5_250',
 'DICT_5X5_50',
 'DICT_6X6_100',
 'DICT_6X6_1000',
 'DICT_6X6_250',
 'DICT_6X6_50',
 'DICT_7X7_100',
 'DICT_7X7_1000',
 'DICT_7X7_250',
 'DICT_7X7_50',
 'DICT_APRILTAG_16H5',
 'DICT_APRILTAG_16h5',
 'DICT_APRILTAG_25H9',
 'DICT_APRILTAG_25h9',
 'DICT_APRILTAG_36H10',
 'DICT_APRILTAG_36H11',
 'DICT_APRILTAG_36h10',
 'DICT_APRILTAG_36h11',
 'DICT_ARUCO_MIP_36H12',
 'DICT_ARUCO_MIP_36h12',
 'DICT_ARUCO_ORIGINAL',
 'DetectorParameters',
 'Dictionary',
 'Dictionary_getBitsFromByteList',
 'Dictionary_getByteListFromBits',
 'GridBoard',
 'RefineParameters',
 '__doc__',
 '__file__',
 '__loader__',
 '__name__',
 '__package__',
 '__path__',
 '__spec__',
 '_native',
 'drawDetectedCornersCharuco',
 'drawDetectedDiamonds',
 'drawDetectedMarkers',
 'extendDictionary',
 'generateImageMarker',
 'getPredefinedDictionary']
```

In [11]:
```
import cv2
import cv2.aruco as aruco
```

```
cap = cv2.VideoCapture('arUco_marker.mp4')
#原始畫面有點大，為了有利於顯示這份講義所以縮小。
frameWidth    = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))//2
frameHeight   = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))//2

arucoParams = aruco.DetectorParameters()#DetectorParameters_create()
arucoDict   = aruco.getPredefinedDictionary(aruco.DICT_7X7_50) #Dictionary_get(a
# Create the ArUco detector
detector = cv2.aruco.ArucoDetector(arucoDict, arucoParams)

print('height {}, width {}'.format(cap.get(cv2.CAP_PROP_FRAME_HEIGHT),cap.get(cv

while True:
    ret, frame = cap.read()
    if not ret:
        break

    frame = cv2.resize(frame,(frameWidth,frameHeight))
    (corners, ids, rejected) = detector.detectMarkers(frame)
    if len(corners) > 0:
        aruco.drawDetectedMarkers(frame, corners, ids)

    cv2.imshow('arUco marker',frame)
    if cv2.waitKey(20) != -1:
        break

cv2.destroyAllWindows()
cap.release()
```

```
height 1080.0, width 1920.0
```

使用未校正相機估計ChArUco Board姿態

相機內部參數先用猜的。

In [12]:

```
import cv2
import cv2.aruco as aruco

cap = cv2.VideoCapture('CharUco_board.mp4')
print('height {}, width {}'.format(cap.get(cv2.CAP_PROP_FRAME_HEIGHT),cap.get(cv
#原始畫面有點大，為了有利於顯示這份講義所以縮小。
totalFrame    = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))
frameWidth    = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))//2
frameHeight   = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))//2

arucoParams = aruco.DetectorParameters()#DetectorParameters_create()
arucoParams.cornerRefinementMethod = aruco.CORNER_REFINE_SUBPIX
arucoDict   = aruco.getPredefinedDictionary(aruco.DICT_6X6_250) #Dictionary_get(
# Create the ArUco detector
detector = aruco.ArucoDetector(arucoDict, arucoParams)

# Create the CharucoBoard
gridX        = 5
gridY        = 7
squareSize   = 4 #cm
charucoBoard = aruco.CharucoBoard((gridX,gridY),squareSize,squareSize/2,arucoDic

cameraMatrixInit = np.array([[ frameWidth/2.,    0., frameWidth/2.],[    0., fra
distCoeffsInit   = np.zeros((5,1))
```

```
charucoParams = aruco.CharucoParameters()
charucoParams.tryRefineMarkers = True
charucoParams.cameraMatrix = cameraMatrixInit
charucoParams.distCoeffs = distCoeffsInit
charucoDetector = aruco.CharucoDetector(charucoBoard, charucoParams,arucoParams)

while True:
    ret, frame = cap.read()
    if not ret:
        break

    frame = cv2.resize(frame,(frameWidth,frameHeight))
    corners, ids, _, _ = charucoDetector.detectBoard(frame)

    if ids is not None and len(ids)>0:
        aruco.drawDetectedCornersCharuco(frame,corners, ids)
        objPoints, imgPoints = charucoBoard.matchImagePoints(corners, ids)
        if objPoints.shape[0]>=4:
            ret, rvect, tvect = cv2.solvePnP(objPoints, imgPoints, cameraMatrixI
            if ret:
                cv2.drawFrameAxes(frame, cameraMatrixInit, distCoeffsInit, rvect

    cv2.imshow('Estimation of the pose of a ChArUco board with inarrurate intrin
    if cv2.waitKey(20) != -1:
        break

cv2.destroyAllWindows()
cap.release()
```

```
height 1080.0, width 1920.0
```

使用ChArUco Board做相機校正

In [13]:

```
import cv2
import cv2.aruco as aruco

cap = cv2.VideoCapture('CharUco_board.mp4')
#原始畫面有點大，為了有利於顯示這份講義所以縮小。
totalFrame   = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))
frameWidth   = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))//2
frameHeight  = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))//2

arucoParams = aruco.DetectorParameters()
arucoParams.cornerRefinementMethod = aruco.CORNER_REFINE_SUBPIX
arucoDict   = aruco.getPredefinedDictionary(aruco.DICT_6X6_250)

# Create the ArUco detector
detector = cv2.aruco.ArucoDetector(arucoDict, arucoParams)

# 必須描述ChArUco board的尺寸規格
gridX        = 5 # 水平方向5格
gridY        = 7 # 垂直方向7格
squareSize   = 4 # 每格為4cmX4cm

# ArUco marker為2cmX2cm
charucoBoard = aruco.CharucoBoard((gridX,gridY),squareSize,squareSize/2,arucoDic

charucoParams = aruco.CharucoParameters()
```

```
charucoParams.tryRefineMarkers = True
charucoParams.cameraMatrix = cameraMatrixInit
charucoParams.distCoeffs = distCoeffsInit
charucoDetector = aruco.CharucoDetector(charucoBoard, charucoParams,arucoParams)

print('height {}, width {}'.format(cap.get(cv2.CAP_PROP_FRAME_HEIGHT),cap.get(cv
refinedStrategy = True
criteria        = (cv2.TERM_CRITERIA_EPS + cv2.TERM_CRITERIA_MAX_ITER, 100, 0.00
frameId         = 0
objPointsList   = []
imgPointsList   = []

while True:
    ret, frame = cap.read()
    if not ret:
        break

    frame = cv2.resize(frame,(frameWidth,frameHeight))
    corners, ids, _, _ = charucoDetector.detectBoard(frame)

    if corners is not None and corners.shape[0]>0:
        aruco.drawDetectedCornersCharuco(frame,corners, ids)
        objPoints, imgPoints = charucoBoard.matchImagePoints(corners, ids)

        if frameId % 100==50 and objPoints.shape[0]>=4:
            objPointsList.append(objPoints)
            imgPointsList.append(imgPoints)

    cv2.imshow('Analysis of a CharUco board for camera calibration',frame)
    if cv2.waitKey(20) != -1:
        break

    frameId += 1

cv2.destroyAllWindows()
cap.release()
```

```
height 1080.0, width 1920.0
```

使用cv2.calibrateCamera 執行相機校正

In [14]:

```
cameraMatrixInit = np.array([[ 1000.,    0., frameWidth/2.],[    0., 1000., fram
distCoeffsInit   = np.zeros((5,1))
ret, charuco_cameraMatrix, charuco_distCoeffs, aruco_rvects, aruco_tvects = cv2.
print(charuco_cameraMatrix)
print(charuco_distCoeffs)
```

```
[[879.94501294   0.         481.92850827]
 [  0.         888.63154407 311.58978267]
 [  0.           0.           1.        ]]
[[ 0.07356283]
 [-0.32777056]
 [-0.00159977]
 [-0.00263169]
 [ 0.86874918]]
```

使用校正的相機估計ChArUco Board姿態

In [15]:
```
import cv2
import cv2.aruco as aruco

cap = cv2.VideoCapture('CharUco_board.mp4')
print('height {}, width {}'.format(cap.get(cv2.CAP_PROP_FRAME_HEIGHT),cap.get(cv

#原始畫面有點大，為了有利於顯示這份講義所以縮小。
totalFrame    = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))
frameWidth    = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))//2
frameHeight   = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))//2

arucoParams = aruco.DetectorParameters()
arucoParams.cornerRefinementMethod = aruco.CORNER_REFINE_SUBPIX
arucoDict   = aruco.getPredefinedDictionary(aruco.DICT_6X6_250) #Dictionary_get(

# Create the ArUco detector
detector = cv2.aruco.ArucoDetector(arucoDict, arucoParams)

gridX        = 5
gridY        = 7
squareSize   = 4 #cm
charucoBoard = aruco.CharucoBoard((gridX,gridY),squareSize,squareSize/2,arucoDic

charucoParams = aruco.CharucoParameters()
charucoParams.tryRefineMarkers = True
charucoParams.cameraMatrix = charuco_cameraMatrix
charucoParams.distCoeffs = charuco_distCoeffs
charucoDetector = aruco.CharucoDetector(charucoBoard, charucoParams,arucoParams)

while True:
    ret, frame = cap.read()
    if not ret:
        break

    frame = cv2.resize(frame,(frameWidth,frameHeight))
    corners, ids, _, _ = charucoDetector.detectBoard(frame)

    if corners is not None and corners.shape[0]>0:
        aruco.drawDetectedCornersCharuco(frame,corners, ids)
        objPoints, imgPoints = charucoBoard.matchImagePoints(corners, ids)
        if objPoints is not None and objPoints.shape[0]>=4:
            ret, rvect, tvect = cv2.solvePnP(objPoints, imgPoints, charuco_camer
            if ret:
                cv2.drawFrameAxes(frame, charuco_cameraMatrix, charuco_distCoeff

    cv2.imshow('Estimation of the pose of a CharUco board with intrinsic camera
    if cv2.waitKey(20) != -1:
        break

cv2.destroyAllWindows()
cap.release()
```

```
height 1080.0, width 1920.0
```

估計Marker姿態

In [16]:
```
import cv2
import cv2.aruco as aruco
```

```
cap = cv2.VideoCapture('arUco_marker.mp4')
markerSize  = 6 #6cm

#原始畫面有點大，為了有利於顯示這份講義所以縮小。
frameWidth    = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))//2
frameHeight   = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))//2

arucoParams = aruco.DetectorParameters()
arucoParams.cornerRefinementMethod = aruco.CORNER_REFINE_SUBPIX
arucoDict    = aruco.getPredefinedDictionary(aruco.DICT_7X7_50)

# Create the ArUco detector
detector = cv2.aruco.ArucoDetector(arucoDict, arucoParams)

print('height {}, width {}'.format(cap.get(cv2.CAP_PROP_FRAME_HEIGHT),cap.get(cv

objPoints = np.array([[0.0,0.0,0.0],[markerSize,0.0,0.0],[markerSize,markerSize,
objPoints = np.array([[markerSize/2,markerSize/2,0],[-markerSize/2,markerSize/2,
while True:
    ret, frame = cap.read()
    if not ret:
        break

    frame = cv2.resize(frame,(frameWidth,frameHeight))
    (corners, ids, rejected) = detector.detectMarkers(frame)
    if len(corners) > 0:
        aruco.drawDetectedMarkers(frame, corners, ids)
        for i in range(len(corners)):
            ret, rvect, tvect = cv2.solvePnP(objPoints, corners[i][0], charuco_c
            cv2.drawFrameAxes(frame, charuco_cameraMatrix, charuco_distCoeffs, r

    cv2.imshow('Estimation of the pose of arUco marker with intrinsic camera par
    if cv2.waitKey(20) != -1:
        break

cv2.destroyAllWindows()
cap.release()
```

```
height 1080.0, width 1920.0
```

世界座標系統$\Rightarrow$影像座標系統

將Marker的4角點透過projectPoints計算其影像座標。

In [17]:

```
import cv2
import cv2.aruco as aruco

cap = cv2.VideoCapture('arUco_marker.mp4')
markerSize  = 6 #6cm

#原始畫面有點大，為了有利於顯示這份講義所以縮小。
frameWidth    = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))//2
frameHeight   = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))//2

arucoParams = aruco.DetectorParameters()
arucoParams.cornerRefinementMethod = aruco.CORNER_REFINE_SUBPIX
arucoDict    = aruco.getPredefinedDictionary(aruco.DICT_7X7_50)

print('height {}, width {}'.format(cap.get(cv2.CAP_PROP_FRAME_HEIGHT),cap.get(cv
```

```
markerCorners3D = np.array([[markerSize/2,markerSize/2,0],[-markerSize/2,markerS

# Create the ArUco detector
detector = cv2.aruco.ArucoDetector(arucoDict, arucoParams)

while True:
    ret, frame = cap.read()
    if not ret:
        break

    frame = cv2.resize(frame,(frameWidth,frameHeight))
    (corners, ids, rejected) = detector.detectMarkers(frame)

    if len(corners) > 0:
        aruco.drawDetectedMarkers(frame, corners, ids)
        for corner in corners:
            for pt in corner[0]:
                cv2.drawMarker(frame,tuple(pt.astype(int).ravel()),(0,0,255),cv2

        for i in range(len(corners)):
            ret, rvect, tvect = cv2.solvePnP(markerCorners3D, corners[i][0], cha
            proj_pt_with_dist,_ = cv2.projectPoints(markerCorners3D, rvect, tvec
            cv2.drawFrameAxes(frame, charuco_cameraMatrix, charuco_distCoeffs, r
            for pt in proj_pt_with_dist:
                cv2.drawMarker(frame,tuple(pt.astype(int).ravel()),(0,255,0),cv2

    cv2.imshow('Estimation of the pose of arUco marker with intrinsic camera par
    if cv2.waitKey(20) != -1:
        break

cv2.destroyAllWindows()
cap.release()
```

```
height 1080.0, width 1920.0
```

In [ ]: